

CSE 280: DIGITAL LOGIC DESIGN

SECTION: 24B

GROUP-1

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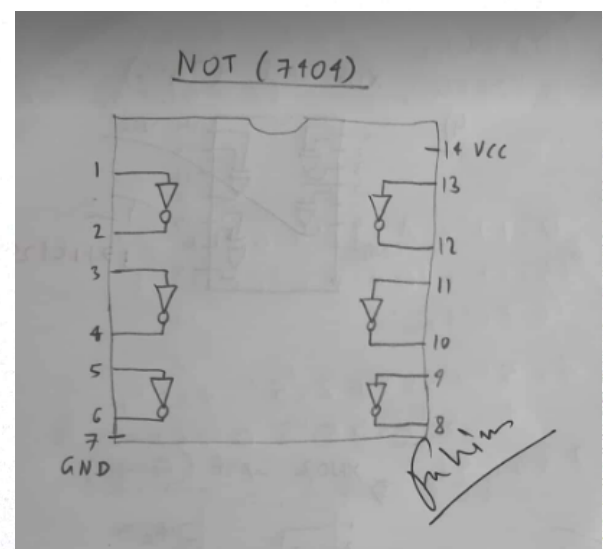
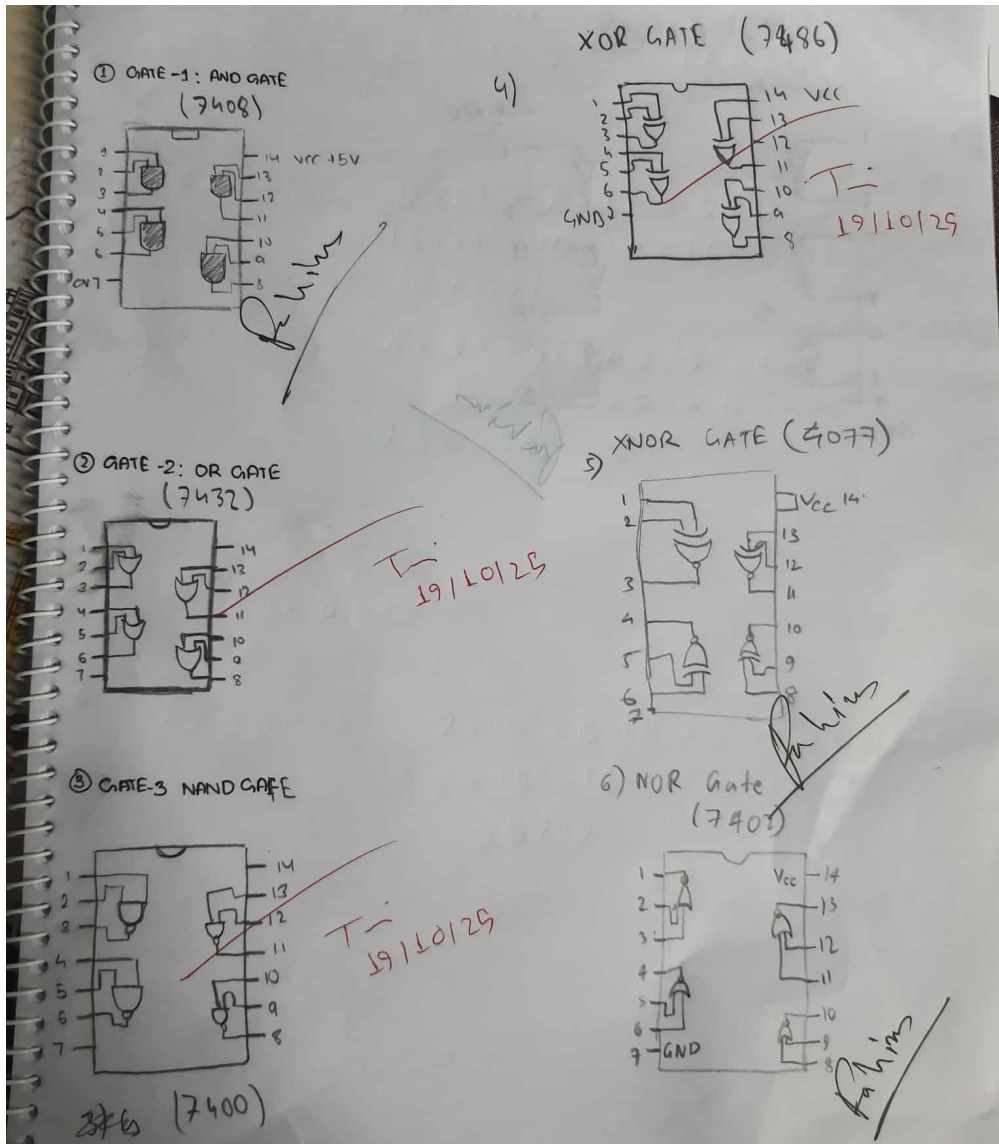
Experiment Name:

- Experiment #1: Familiarization with fundamental logic Gates.
- Experiment #2: Universal Gates, Application of Boolean Algebra
- Experiment #3: Parity Bit Checker & Generator.

Required Components for Experiment - 1, 2, 3 :

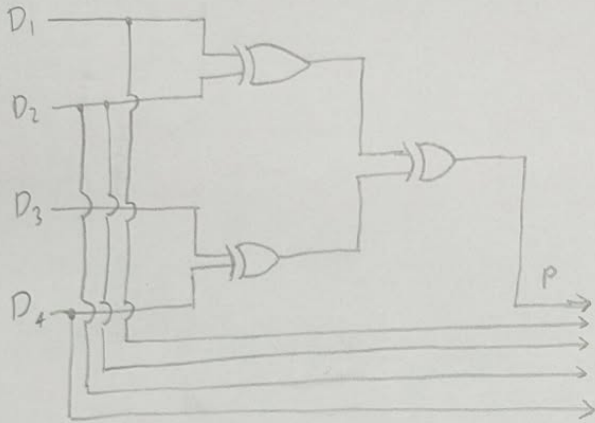
- 1) IC 7408 x 1
- 2) IC 7432 x 1
- 3) IC 7404 x 1
- 4) IC 7400 x 1
- 5) IC 7402 x 2
- 6) IC 7486 x 1
- 7) IC 4077 x 1

• Experimental Setup & Result for Experiment 1.



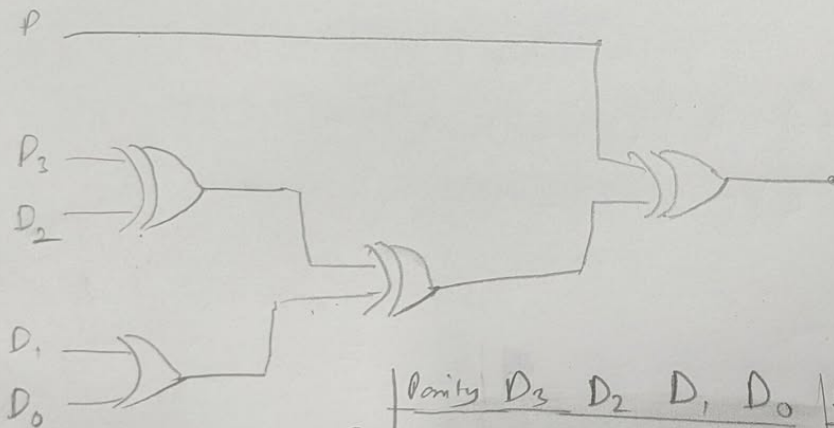
• Experimental setup & Result for experiment 3:

Even Parity Generator



	Data				Parity
	D ₃	D ₂	D ₁	D ₀	
a.	1	0	0	1	0
b.	0	0	0	1	1
c.	1	1	1	1	0
d.	0	0	0	0	0

Pravin
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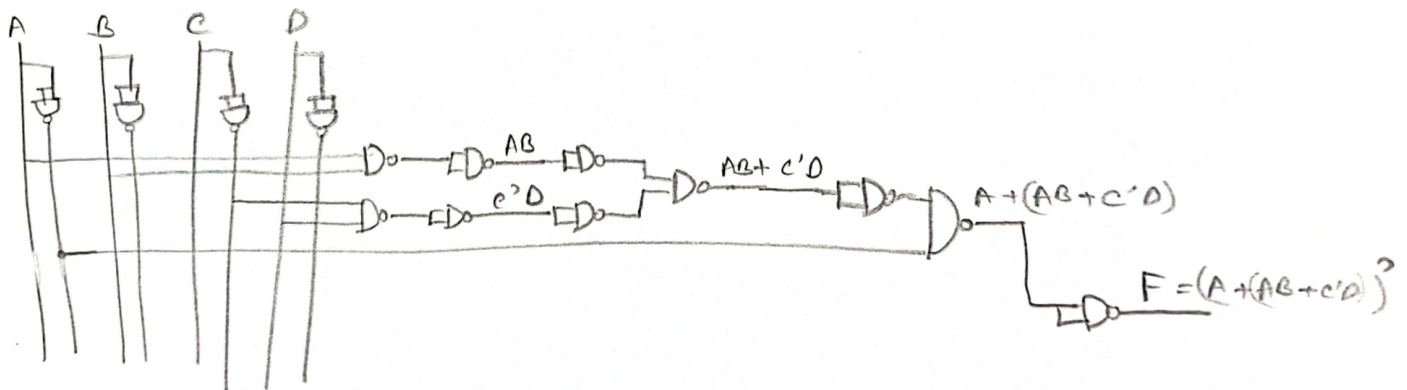


	Parity	D ₃	D ₂	D ₁	D ₀	Error
a	1	1	0	0	1	1
b	0	0	0	0	1	1 ✓
c	0	1	1	1	1	0 ✓
d	1	0	0	0	0	1 ✓

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Implement the following function using NAND gate only:
 $(A + (ABc'D))'$. Do not simplify the function. Draw the
 Circuit diagram only.

1) $F = (A + (AB + c'D))'$



• Draw the circuit diagrams of 3-bit parity checker & generator using NOR gates only.

b) 3 bit parity even generator with NOR gates only is given below: -

$$F = A \oplus B \oplus C$$

$$= (A \bar{B} + \bar{A} B) \oplus C$$

$$= \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} \bar{C} + A B C$$

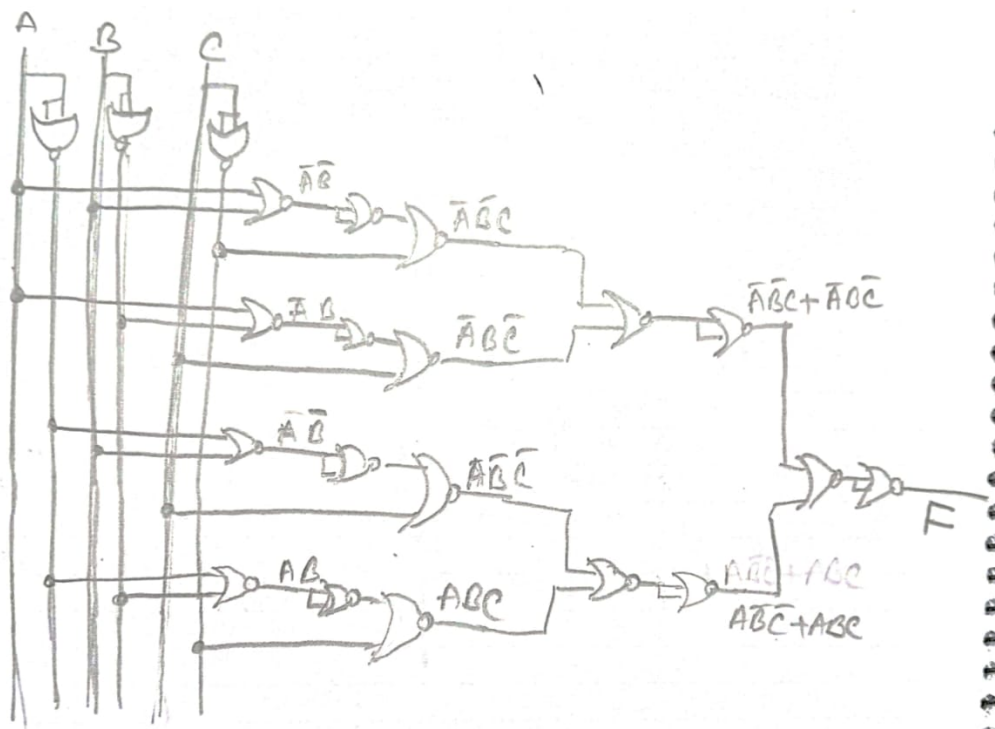


Fig: Generator

